

Project Initialization and Planning Phase

Date	28 June 2026
Team ID	SWTID-2026-9068
Project Name	Smart Lender
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective of Smart Lender is to automate the loan approval process using Machine Learning. It evaluates customer eligibility based on financial information and predicts whether a loan should be approved accurately and quickly.
Scope	The project collects applicant details, preprocesses data, predicts loan approval using an ML model, stores application records, and displays the result through a simple web interface.
Problem Statement	
Description	Traditional loan approval systems are time-consuming, involve manual verification, and may lead to errors or biased decisions. Customers often experience delays in receiving loan decisions.
Impact	Smart Lender reduces processing time, improves prediction accuracy, minimizes human errors, and provides faster and more reliable loan approval decisions for both banks and customers.
Proposed Solution	

Approach	Use Machine Learning algorithms with Python and Streamlit to analyze applicant information, predict loan eligibility, and provide instant loan approval recommendations
Key Features	• Loan eligibility prediction using ML. • User-friendly web interface. • Fast loan approval decision. • Secure customer data handling. • Real-time prediction results. • Stores applicant details for future reference.



Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU	Intel Core i5 or above
Memory	RAM	8 GB or higher
Storage	Disk space	256 GB SSD
Software		
Frameworks	Python frameworks	Streamlit , Scikit-learn
Libraries	ML libraries	Pandas, NumPy, Scikit-learn, Joblib, Matplotlib
Development Environment	IDE	Visual Studio Code
Data		

Data	Source, size, format	Loan applicant dataset (CSV), including applicant income, loan amount, education, credit history, employment status, and loan status for training and prediction.
------	----------------------	---